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Research

Influences of international trade policy and the COVID-19 Pandemic on fishery industry: an example of the Taiwan hairtail industry

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Abstract

Sustainability in fisheries is critical to the global food supply, making the study of fishery industries essential. Taiwan's hairtail fishery represents a significant component of the nation's fishing industry, offering valuable insights into broader trends and challenges. This study aims to examine the influences of changes in trade policy and trade conditions (COVID-19 pandemic) using the hairtail industry in Taiwan as an example, providing insights into how external shocks, such as global health crises and policy reforms, affect the fisheries sector, particularly in regions reliant on small-scale or traditional fishing practices. The findings indicate that hairtail productions and prices remained stable between 2012 and 2015. However, increasing demand and export opportunities in China drove significant growth in production and market prices from 2016 to 2019. This upward trend was abruptly disrupted in 2020 by the COVID-19 pandemic, which caused a sharp decline in production and prices. The return of unsold products from the Chinese market to the local market further suppressed prices, highlighting the vulnerability of the industry to external shocks. Digitalization played a crucial role during this period, facilitating the even redistribution of products in domestic markets, stabilizing prices, and reducing losses. Looking forward, further digitalization is essential to enhance the resilience of the hairtail fishery. Advanced digital tools will not only strengthen the industry's ability to adapt to future disruptions but also ensure traceability, meeting the demands of international trade and fostering sustainable growth.

Highlights

- Decadal hairtail production and values were studied under the influences of increasing international demand and COVID-19 pandemic.
- How the hairtail industry in Taiwan addressed challenges through strategy and digital approaches was first studied.
- Further development and uses of digital transformation in the marine fishery industry are discussed and proposed.

Keywords COVID-19 · Digital Transformation of Industries · Fishery · Marketing Channel · Hairtail · Production Management · Taiwan · SDG

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1 Introduction

Fisheries products are important sources of animal protein and omega-3 fatty acids [2]. Food from its industry contributes actively to food security for the world's growing population. Generally, the fishery industry includes production, processing, and marketing, which are highly influenced by the changes in global fishery resources, global fishery product trade system, sudden changes in the fishery product trade market, and changes in consumer preferences [11–14]. Understanding the influences of various policy and environmental factors on fishery industry is essential to achieve sustainable development of fisheries.

Since 2020, the global COVID-19 pandemic has dramatically hindered international trade, causing supply chain disruptions, delivery delays, and increased transportation costs. As such, the global economy, trade system, and consumer consumption behaviours have gradually changed [10]. In response to the COVID-19 pandemic, since 2021, the rise of the stay-at-home economy has significantly changed the international trade market, including the fishery products market [17]. Therefore, many countries have begun to develop stay-at-home economy and the relevant trade policies. The feasibility of introducing new commercial management models into the global trade market is widely discussed to strengthen the resilience of the overall fishery products market and to reduce the direct or indirect impact on the industry caused by changes in the external environment [17, 19]. The resilience used in this study refers to the capacity of the industry to anticipate and reduce, cope with, and respond to and recover from external disruptions based on the definition of the Intergovernmental Panel of Climate Change [16].

The study of the influence of the COVID-19 pandemic is essential in understanding the global fishery sector because it highlights the vulnerabilities and resilience of this critical industry during unprecedented global disruptions. Fisheries are a cornerstone of global food security, livelihoods, and trade, and the pandemic exposed how interconnected and fragile these systems can be. However, a common phenomenon is that there is an insufficient number of reliable and continuous records for these processes, which makes it challenging to gather essential information.

Consequently, the seafood product industry faces a higher risk and lower resilience due to low data integrity and limited public availability, particularly in response to significant changes in external factors like international trade patterns and seafood resource availability. Low data integrity and public availability, however, cause the seafood product industry to have higher risk but lower resilience, especially under drastic changes in external factors such as international trade patterns and seafood resources [10]. The decadal open-access record of the fishery industry from the Fisheries Agency of Taiwan provides an invaluable resource to study the impacts of various factors on the fishery industry.

Taiwan is one of the world's leading countries in fishery, with its fleet operating in international waters across the Pacific, Indian, and Atlantic Oceans. Taiwan is a major exporter of seafood products. It exports a variety of seafood, including tuna, squid, and processed fish products, to markets like Japan, the United States, and the European Union. It also imports seafood to meet domestic demand and for re-export. While Taiwan may not be the largest producer in terms of volume, its contributions to high-value fisheries, technological advancements, and global trade makes it a key player in the world fishery industry.

This study aims to examine the influences of changes in trade policy and trade conditions (COVID-19 pandemic) using the hairtail industry in Taiwan as an example. Hairtail is a representative segment of the country's fishery market due to its economic significance, cultural relevance, and reflection of broader industry dynamics. Hairtail is one of Taiwan's most commercially important fish species, widely consumed domestically and exported to international markets, showcasing the dual reliance of Taiwan's fisheries on local and global demand (Fig. 1). The industry embodies key characteristics of Taiwan's fishery sector, including small-scale fishing practices, regional market integration, and challenges related to resource sustainability. Examining this industry offers valuable insights into how external shocks, such as global health crises and policy reforms, affect the fisheries sector, particularly in regions reliant on small-scale or traditional fishing practices.

Firstly, the hairtail industry in Taiwan is emblematic of challenges faced by fisheries worldwide. Policy changes, such as stricter environmental regulations, resource management strategies, and trade policies, can significantly impact fishing practices, economic viability, and resource sustainability. Understanding how Taiwan's hairtail fishery adapts to these changes provides a case study for balancing conservation with economic growth—an issue faced by fisheries globally.

Secondly, the COVID-19 pandemic has underscored the vulnerability of global supply chains and the resilience of local industries. In Taiwan, disruptions to labor, transportation, and market demand during the pandemic highlighted

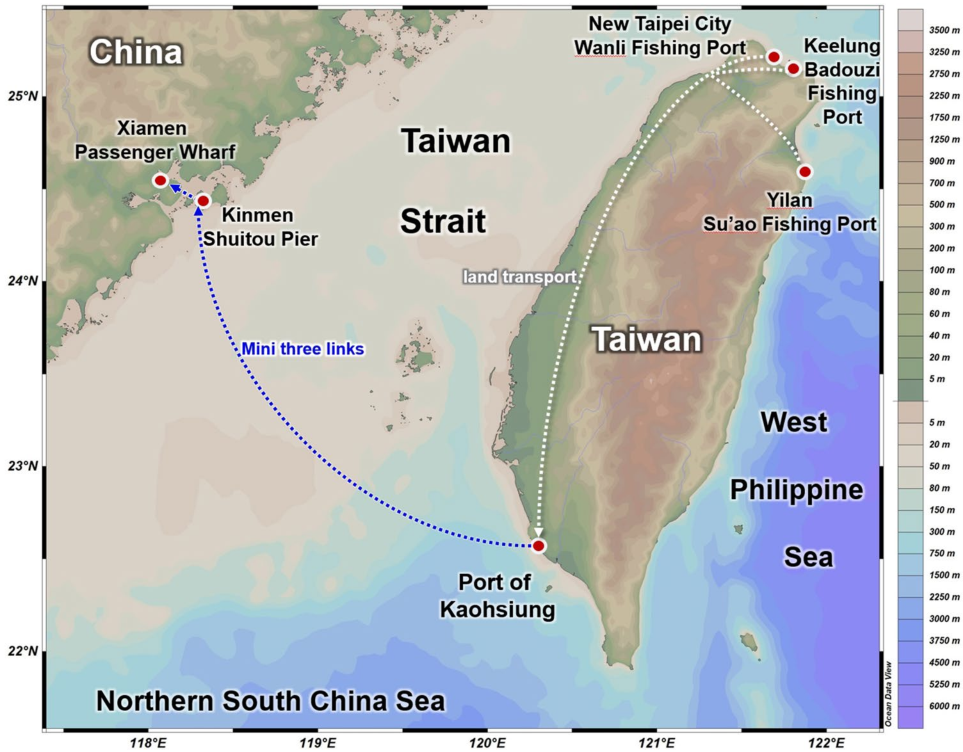

Fig. 1 The transportation route for Taiwan's hairtail fishery exports to China via the Mini Three Links

the fragility of fisheries dependent on both domestic and export markets. By studying how the hairtail industry navigated these disruptions, researchers can derive lessons on building resilience in the face of global crises. These insights are particularly relevant for other countries with similar fisheries-dependent economies.

Thirdly, Taiwan's position as a significant player in the Asia–Pacific fishery sector amplifies the relevance of this study. The region accounts for a substantial portion of global fishery production, and understanding Taiwan's experience contributes to a broader understanding of how regional policies and global events shape fisheries worldwide. This research not only informs sustainable fisheries management but also aids in developing strategies to mitigate the impacts of future disruptions on global food systems.

2 Materials and methods

2.1 Datasets of hairtail industry in Taiwan

Data of the annual hairtail production between 2021 and 2022 were taken from the "Fisheries Statistical Yearbook Taiwan, Kinmen and Matsu Area" published by the Fisheries Agency, Ministry of Agriculture, Taiwan [7]. Data of the trading volume and price of the hairtail fishery products between 2013 and 2022 were taken from the Fishery Products Global Information Website [8]. Data of the export volume of hairtail fish products to China in 2013–2022 were taken from the "Monthly Report on Customs Import and Export Trade Statistics" published by the Customs Administration Ministry of Finance [3].

2.2 Stakeholder composition and implementation method of in-depth expert interviews

To achieve diversified opinion collection, this study developed a topic outline in advance and conducted in-depth expert interviews with key stakeholders to understand stakeholders' opinions on the current situation of the hairtail fishery industry, the operational problems faced, and the possible opportunities related to these risks. Feedback recorded on paper included the current management and production status, import and export structure and proportion, market model and channel, COVID-19 pandemic issues, whether to diversify risk and the reasons, views and acceptance regarding the introduction of a commercialization model and digital transformation which includes digitalization and digital optimization.

2.3 Identification strategy and the composition of stakeholders

To capture the situation of various stakeholders on the production, transportation and marketing of hairtails in Taiwan, field surveys were implemented mainly in domestic loading ports designated by the "Procedures for the Production and Marketing of Hairtails along the Coast and the Operation of Fishing Boats". To understand the current situation, problems and possible future opportunities of the hairtail fishery industry, this study adopted the "principle of stakeholder identification" proposed by the United Nations Human Settlements Programme (UN-Habitat) in 2014 and 2020 as the main criteria for identification. During policy formulation and stakeholder identification, a systematic approach was used to target and assist in identifying stakeholders and to help converge on the research topics. Additionally, this study employed a snowball sampling method, continuously asking for suitable or subsequent recommendations for other potential key stakeholders during the on-site surveys (direct observations and records) and in-depth interviews (with questionnaires shown in 2.4) to continually make changes or add key stakeholders (UN-Habitat 2014 & 2020). Based on the results of the stakeholder identification, the interviewees were mainly stakeholders such as fishery operators, suppliers, fishermen's associates, the central government and local governments (Table 1). There were 60 stakeholders involved in the in-depth interviews. The respondents included fishery management units (10 people), merchant middlemen (13 people), dealers at various sales stages (7 people), fishery operators (30 people), underwriters and local industry practitioners.

2.4 In-depth expert interview questions

The key points of the in-depth interviews with experts included several items:

A. What are the differences in the sales structure and catch between the "mini three links" and "direct shipping and sale by carrying vessels" channels?
B. Has the COVID-19 pandemic changed and impacted the global trading economy and the transportation and marketing of hairtail production and export? What is the extent of the impact and shock?
C. Under the influence of the COVID-19 pandemic, has the policy implementation or the transportation and marketing conditions of the Regulations for Managing the Production and Marketing of Coastal Hairtail and Fishing Vessels Carrying Coastal Hairtail been promoted by the Fishery Administration?
D. Regarding the introduction of commercialized management models (such as technology digitization and digital transformation) into the fishing industry, what are the industry's level of acceptance, views and opinions? What potential problems or bottlenecks may be encountered in the commercialization model? How can potential problems be addressed to create future opportunities for the industry?

2.5 MAXQDA qualitative analysis

The transcribed interviews from stakeholders of the hairtail fishery industry were uploaded to the MAXQDA qualitative data organization software program, utilising the Grounded Theory to synthesize the data into a theory (Strauss, A., & Corbin, J., 1998) to analyse hairtail fishery industries' adaptation measures in response to COVID-19, identify causal relationships between the research issues, summarize existing digitalization actions, systematically compile a considerable amount of field data, and derive theoretically grounded implications.

3 Results

3.1 The production, supply and demand trends before the implementation of hairtail fishery management policies in 2019

Figure 2 shows the annual production and value of the hairtail fisheries in Taiwan between 2012 and 2022. The trends of the value generally followed that of the annual production. The annual production of hairtail was 5471 tons in 2012 and showed no significant variations to 2015 (indicated with "stable stage" in Fig. 2), having an average of 4761 ± 576 tons between 2012 and 2015. The annual productions increased rapidly from 6,012 tons in 2016

Table 1 List of stakeholders in the definition of material flow

Corresponding factor	Livelihood issues	Operational risk changes	Environmental change	Uncertainty in fishery trade supply and market supply
Corresponding unit	Fishery operators, merchant middlemen, dealers at various sales stages	Fishery operators	Fishery operators, fishery management units	Fishery management units, fishery operators, merchant middlemen, dealers at various sales stages

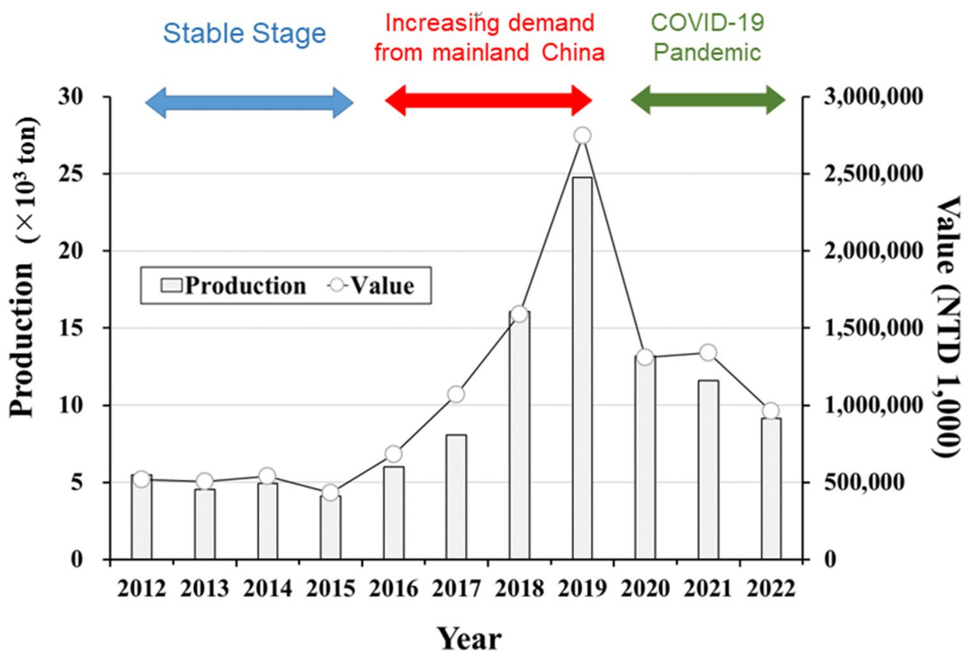

Fig. 2 Changes in the total production and value of hairtail fisheries in Taiwan from 2012 to 2022. (*Source*: Fisheries Agency, Council of Agriculture, Executive Yuan (2022a)). NT 1 is about US 0.03

to the highest of 24,743 tons in 2019, which was 420% increment when compared with the average between 2012 and 2015. A sudden decrease of 47% in the production occurred in 2020 when compared with that in 2019, and the following were slightly decreases in 2021 and 2022.

Trends of the market value generally mimicked that of the annual production. The annual value showed in significant variation between 2012 and 2015, having an average of NT 499,961 ± 46,189 thousand (NT 1 is about US 0.03). The value increased in 2016 of NT 681,012 thousand to the highest of NT 2,747,549 thousand in 2019, which was 450% increment when compared with the average between 2012 and 2015. Similar to the annual production, a sudden decrease of 52% in the market value occurred in 2020 when compared with that in 2019, and the following were slightly decreases in 2021 and 2022. Indeed, the increases in both the annual production and values between 2016 and 2019 were due to the expand on the demand from mainland China, whereas the sudden drops stared in 2020 were due to the impact from the COVID-19 pandemic, and discuss as follows.

Due to the increasing demand for hairtails from mainland China, the export of hairtails in Taiwan has gradually expanded in 2016−2019. Figure 3 shows the export of hairtail products to mainland China. The data showed that there was almost no export to mainland China in 2013−2015. Similar to the annual production shown in Fig. 2, the export began to increase from 2,153 tons in 2016 to the highest of 24,622 tons in 2019, and then decreased gradually since 2020 of 19,368 tons to 3,682 tons in 2022 (Fig. 3). Worth mentioning is that the annual production and price information shown in Fig. 2 and the export information shown in Fig. 3 came from different sources. The magnitudes of changes in the export production matched well that of the annual production, suggesting that the changes in the annual production and value largely were due to the changes in the export production. Again, the increases in export between 2016 and 2019 were due to the increasing demand from mainland China, and the decreases started from 2020 were due to the COVID-19 pandemic. Certainly, the price also was affected. From 2013 to 2015, the amount of hairtail traded in the wholesale market of Taiwan (domestic hairtail consumption) was approximately 1345 tons, and the average price was approximately 146.6 NT/kg (Table 2, [8]). After the increased export of hairtails to China during 2016–2018, the wholesale market hairtail trade was on average 1,379 tons, and had an average price of 163.3 NT/kg. The price increased by approximately 11.6% (Table 2,[8]). During the COVID-19 pandemic, the price reduced to a similar values of 150.9–151.5 NT/kg in 2019–2021, and reached the lowest price of 142.7 NT/kg in 2022. The reduction in the prices of the domestic market during the pandemic was because of the increasing supply of hairtails, as catches could not be exported to China were eventually returned to the domestic fishery market.

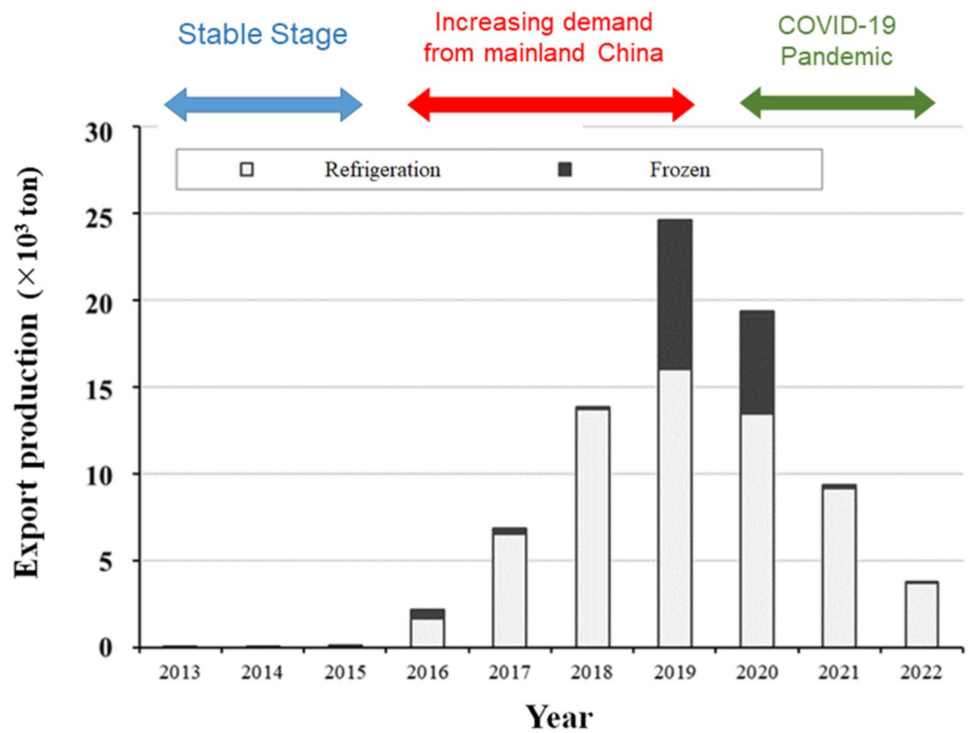

Fig. 3 Statistics of exports of hairtail products to mainland China from 2013 to 2022. (*Data source*: Monthly Report on Customs Import and Export Trade Statistics, 2022)

Table 2 Domestic market trading volume and average price of hairtail products from 2013 to 2022

Year	Volume (ton)	Price (NT/kg)	Price increase ratio (%)
2013	1,349	151.7	–
2014	1,433	141.1	− 6.99
2015	1,252	147.1	4.25
2013–2015 Average	1,345	146.6	
2016	1,197	160.4	9.04
2017	1,486	170.0	5.99
2018	1,455	159.5	− 6.18
2016–2018 Average	1,379	163.3	
2019	1,390	150.9	− 5.39
2020	1,609	151.5	0.40
2021	1,446	150.1	− 0.92
2019–2021 Average	1,482	150.8	
2022	1,510	142.7	− 4.93

Source: Fisheries Agency, Council of Agriculture, Executive Yuan (2022b)

3.2 Impacts of COVID-19 pandemic on a new open trading policy and the market in late 2019

Before 2019, catches were transported to Kaohsiung City through land transport and then transported to Kinmen via the Port of Kaohsiung for customs clearance. The catches were then transported to Xiamen City, China. The route from Kinmen to Xiamen City is so-called "mini three links" in the industry (Fig. 1). In response to the demand for industry operators to expand the hairtail industry in mainland China and reduce operation costs, on July 24, 2019, Taiwan's Fishery Administration approved the "Regulations for Governing the Trial Operation of Fishing Vessels Carrying Coastal Hairtails" [6]. The legislative process and the content of the regulations are detailed in Table 3

Table 3 The history of fisheries management policy development in the hairtail fishery industry

Year	Operation and sales model and policy management change in hairtail fishery industry
Before 2016	The sales market was the domestic consumer market in Taiwan; after fish is caught, it can be directly stored in an ice bucket
After 2016	Due to the increase in the demand for hairtails by the mainland China seafood product consumption market, the export market for hairtails from Taiwan gradually expanded, and the production, output value and export volume of hairtail fisheries in the coastal waters in Taiwan began to increase in 2016
July 24, 2019	Taiwan's Fishery Administration approved the "Regulations for Governing the Trial Operation of Fishing Vessels Carrying Coastal Hairtail"
From July 24, 2019, to October 2019	Industry operators reported that the mode of transportation and sales was limited by the long delivery times, which made it difficult to preserve the freshness of the catch and lowered the market sales price. Therefore, industry operators looked forward to Taiwan's fishing export policy to open the path for selling fresh hairtail to the Chinese market by fishing boats
October 7, 2019	Taiwan's Fishery Administration approved the "Regulations for Managing the Production and Marketing of Coastal Hairtails and Fishing Vessels Carrying Coastal Hairtails"
	This management regulation defined the management terms and objects of the hairtail fishery industry, including hairtail transportation vessels, ship groups, regulations for completing the declaration of fish unloading, designated domestic loading ports for the transport of hairtail, transportation and navigation regulations, etc
December 2019	The COVID-19 pandemic began in December 2019. Following changes in China's border control and port management regulations, the local city lockdown and the reduction in the proportion of consumers leaving, the amount of hairtail catches transported to China after 2020 decreased
	The "Regulations for Governing the Trial Operation of Fishing Vessels Carrying Coastal Hairtail" were suspended
2020–2021	The export supply chain faced problems such as prolonged shipping schedules, PCR testing, declining catch freshness and changes in market and consumer consumption patterns, resulting in a high time cost for catch to be sold to the Chinese market and affecting the freshness and price of fish
	Operators and dealers began to adjust the production and sales model of hairtail in the Taiwanese and Chinese markets and the ratio of catch distribution, resulting in the phenomenon that catches that could not be sold to the Chinese market began to flow back to the Taiwanese consumer market
	The method of transporting fish catches by freight and use the "mini three links" route to directly transport and sell fish catches from Taipei or Kaohsiung ports to the China market were implemented
	The industry attempted to establish a financial digital diary of fishery products trade through smartphones to support market decision-making
	Using the real-time conditions of different sales markets, this technology was used to determine customer preferences and allocate different specifications of catches and market allocations in real time to reduce unsellable catches and possible business losses

Source: Fisheries Agency, Council of Agriculture, Executive Yuan (2019) and compilation of in-depth interviews with stakeholders

and Table 4, respectively. In addition to the "mini three links" route, this regulation allows for hairtail transportation vessels to directly transport refrigerated hairtails to the Chinese market (Table 3 & Table 4). Under regulation, if the industry operator has obtained an operating permit for hairtail transport boats, the transport boats will be able to collect and carry the hairtail caught by fishing boats and sail to a Chinese port for offloading, and there is no limit on the number of boats.

The regulation effectively reduces the transportation time and cost in the industry and could result in increased sales of hairtail caught off Taiwan's coast to the mainland China market. Unfortunately, the COVID-19 pandemic began in December 2019. Following changes in China's border control and port management regulations, local city lockdowns and reductions in consumption, the number of hairtail catches transported to China after 2020 significantly decreased.

The main impacts of the COVID-19 pandemic on the hairtail industry are the time cost of fishery activities and catching, transportation, and marketing. Following the COVID-19 outbreak beginning in December 2019, Taiwan's hairtail export supply chain faced problems such as prolonged shipping schedules, PCR testing, declining catch freshness and changes in market and consumer consumption patterns. These problems resulted in a high time cost for the catch to be sold, affecting the freshness and price of catches. However, due to the lack of information and opacity in the production and distribution markets of the hairtail industry in the past, the industry has had difficulty achieving resilience. Consequently, the export amount decreased in 2020 (Fig. 3). As mentioned, catches that could not be exported to China were returned to the domestic fishery market, reducing the price of hairtail in the domestic market. During the COVID-19 pandemic, the mainland China required addition information for the

Table 4 Overview of hairtail fishery industry guidance and catch export management measures

Regulations for Managing the Production and Marketing of Coastal Hairtail and Fishing Vessels Carrying Coastal Hairtail	
Area of regulation	Content description
Hairtail transportation vessels	According to the provisions of the regulations, a permit is required to use fishing vessels for transporting hairtail along the coast
Ship group	Refers to the cooperation of a fixed fishery, specific fishing vessels and transport vessels for hairtail in coastal waters
Regulations for completing the declaration of fish unloading	A declaration of fish unloading along the coastal waters is required for fishing groups of hairtail fishing boats
Designated domestic loading port for the transport of hairtail	A declaration of fish unloading along the coastal waters must be completed by any fishing group of hairtail fishing boats
	Fishermen who transport hairtail should provide the loading and unloading list for each voyage in the previous month, the declaration of fish unloading from the group of fishing vessels and the certificate of fish trade to the central competent authority before the tenth of each month
	Badouzi of Keelung City, Aodi of New Taipei City, Yehliu City, Wushi of Yilan County, Nanfang'ao, Hsinchu City of Hsinchu City and Wuqi Fishing Port of Taichung City are the designated ports. However, due to the impact of the COVID-19 pandemic, Taiwan's Central Epidemic Command Center has declared mainland China as a travel epidemic recommended level 3 warning area, and the Hsinchu fishing port in Hsinchu City is now the designated port for transportation
Transportation and navigation regulations	Hairtail transportation vessels that transport hairtail outside Taiwan are limited to ports in the mainland region that are approved by the management regulations for this region for fishing vessels to sail to the mainland
	When hairtail transportation vessels return from the mainland, they are not allowed to carry any items
	When hairtail transportation vessels return from the mainland, they should sail directly to the original fishing port to enter the port and undergo an inspection by the maritime patrol agency. During the voyage, they are not allowed to dock or stop at the box net aquaculture site at sea

Source: Fisheries Agency, Council of Agriculture, Executive Yuan (2019)

import of hairtail. The main reasons for import prohibition were (1) a lack of data on production operations, marketing, and research; (2) the absence of clear official information on catch volume and transaction volume; (3) a lack of information about the supply and marketing chain of the industry; (4) the lack of assessment of potential risks in the supply chain; and (5) the potential impact of fluctuations in the export market on supply and demand in the domestic market. Indeed, this is the trend of change for the international trade in fishery products that importers request for traceable products, which need to record information in detail for every process of the industry.

3.3 On-site survey and interview: problems, challenges, transition and achievements

3.3.1 Domestic market

According to the on-site survey, hairtails are sold mainly fresh or refrigerated for export or transported by dealers to the domestic seafood product trading market for sale. The on-site survey and interviews indicate that the domestic consumer market sales channels for hairtail catches in northern Taiwan can be divided into two main types: (1) After the fishing boats enter the port, the catches are sold to dealers in the form of fresh fishery products. After being purchased by a specific dealer, the products are stored in seawater ice and classified as refrigerated fishery products. They are then transported to the wholesale fishery market by vehicles for sale. (2) The operators sell fresh fishery products to restaurants through local produce channels or to consumers in consumer markets (Fig. 4).

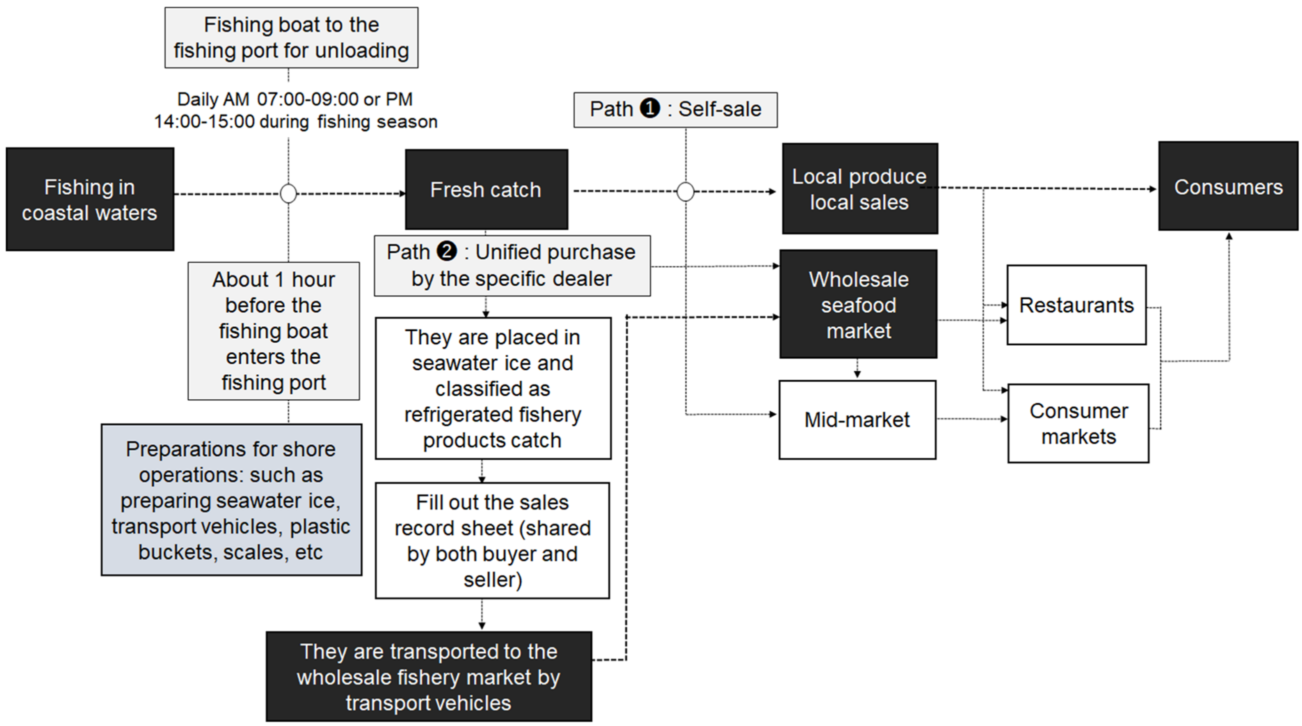

Fig. 4 Marketing channels of the hairtail fishery industry in the domestic consumer market. Boxes with background in black represent the main processes of production and sales, the gray ones provide transportation information, the white ones indicate transit stations or processes, and the light blue one shows the preservation methods used by producers during the transport of seafood to designated ports

3.3.2 Overseas market

In terms of the structure of overseas consumer market, before the "Regulations for Governing the Trial Operation of Fishing Vessels Carrying Coastal Hairtail" policy, the export model of hairtail fish in China was based mainly on the so-called “mini three links” (Table 3): After being caught by Taiwanese fishing vessels along the offshore waters, the catches are transported to the fishing ports designated by the management unit to trade with dealers (including Badouzi Fishing Port in Keelung, Wanli Fishing Port in New Taipei City and Su’ao Fishing Port in Yilan) [6]. After the transaction is completed, the catch is transported to Kaohsiung City through land transport by packing in 20-foot freezer containers, and then transported to Kinmen via the Port of Kaohsiung for customs clearance. The catches are then transported to Xiamen City, China, through the so-called “mini three links” transportation and sales channel (Fig. 1). After the customs duty is paid, catches enter China’s seafood market. The overall transportation and marketing channel operation time is approximately 2–3 days (see Fig. 5) [6]. The long delivery time makes it difficult to preserve the freshness of the catch and lowers the market sales price. Therefore, industry operators look forward to Taiwan’s fishing export policy to open a new way to sell fresh hairtail directly through fishing boats to the Chinese market.

3.3.3 The impact of the COVID-19 pandemic on hairtail fishery industry overseas sales

Due to the COVID-19 pandemic and the border management measures implemented by the Taiwanese government, foreign fishermen working on fishing vessels and fish-borne vessels could not enter Taiwan. Thus, many fishing vessels cannot be operated due to the lack of crew. Meanwhile, the labour force invested in the industry decreased. In addition, relevant inspection and quarantine must be performed after each sea trip. The time required to invest in the inspection causes an operation time delay, and the inspection cost increases the fishery operator’s cost. At the same time, due to COVID-19 pandemic control measures and to avoid the risk of infection, catches must be placed in ports for 2 days after being sent to the Chinese market before landing. After landing, PCR analysis of the catches was performed to confirm that they did not contain viruses before they entered the market. Taking the “mini three links” channel as an example, if the catch is shipped from the Port of Kaohsiung every Wednesday and in a normal condition, the catch will arrive on Saturday. However, the catch could only land on the following Monday, as the market in Xiamen city is closed on Sunday.

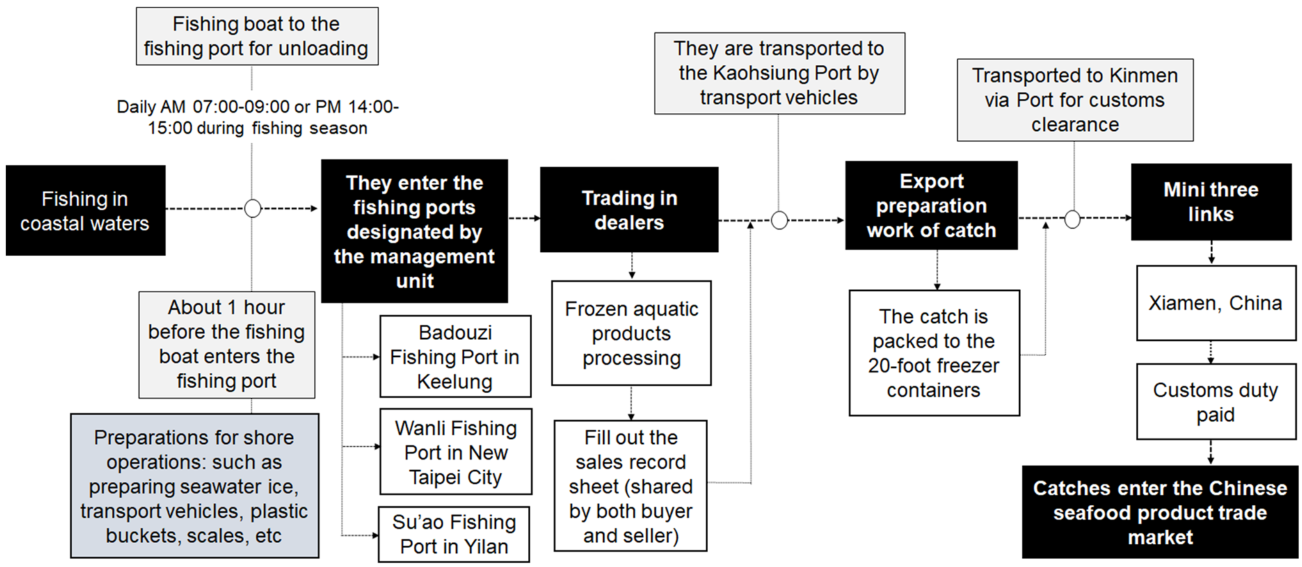

Fig. 5 Transportation and sales channels of the hairtail fishery industry in the export consumer market. Boxes with background in black represent the main processes of production and sales, the gray ones provide transportation information, the white ones indicate transit stations or processes, and the light blue one shows the preservation methods used by producers during the transport of seafood to designated ports

Thus, it takes six days for the delivery. In the case of the pandemic, it took an additional two weeks to complete tests and several procedures.

According to the stakeholder interviews, the industry initiated joint production. To ensure consistency in information transmission and enhance information transparency, the industry entrusted related companies to create electronic catch record logs and information exchange platforms. Therefore, the industry can avoid the uneven distribution of fish catches in markets. This technology also allows for stakeholders to track the sources of fishery products, sales routes, production conditions, and current trading pricing in fishery markets. Using the real-time conditions of different sales markets, this technology can help determine customer preferences and allocate different specifications of catches and market allocations in real time to reduce unsellable catches and possible business losses.

3.4 Analytical results of the grounded theory of MAXQDA

According to the coding proportion of the MAXQDA analysis, the operators of the fishery production end consisted of 49%, 21% was middlemen, 13% was seafood traders, 13% was domestic market distributors, and 4% was fishery authorities. After identifying the key targets in the industry, the interviewees were narrowed down to fishery operators. Based on the analytical results of Grounded Theory, the preconditions for the stable production and sales of the hairtail fishery industry in Taiwan include COVID-19 (46%), policy changes in hairtail fish exports to the Chinese market (19%), and disruptions in the Chinese market and border control measures (35%), resulting in decreased stability in the production and sales chains of the hairtail fish industry (Fig. 6).

To cope with the changes in production and sales patterns caused by COVID-19, as well as the resulting "stay-at-home economy" consumption pattern, stakeholders have adopted action/interaction strategies, which include developing processing of byproducts and transporting frozen catches (9%), increasing domestic market sales and channels (39%), reducing operational costs through fishery subsidies (5%), developing other international trade markets (3%), increasing the number of trade container days for export to China (10%), establishing a mobile digital financial catch log system (21%), and introducing digital technology to grasp real-time sales conditions in the market (13%). However, contextual conditions that influence the actions taken include place costs (12%), the freshness and price of catches (63%), and changes in consumption patterns (25%). In addition, the intervening conditions include Taiwan's market allocation (62%), the mini three-link route (22%), and the development of international trade markets (8%). Together, these factors increase the stability of production and sales in Taiwan's hairtail fishery industry (Fig. 6).

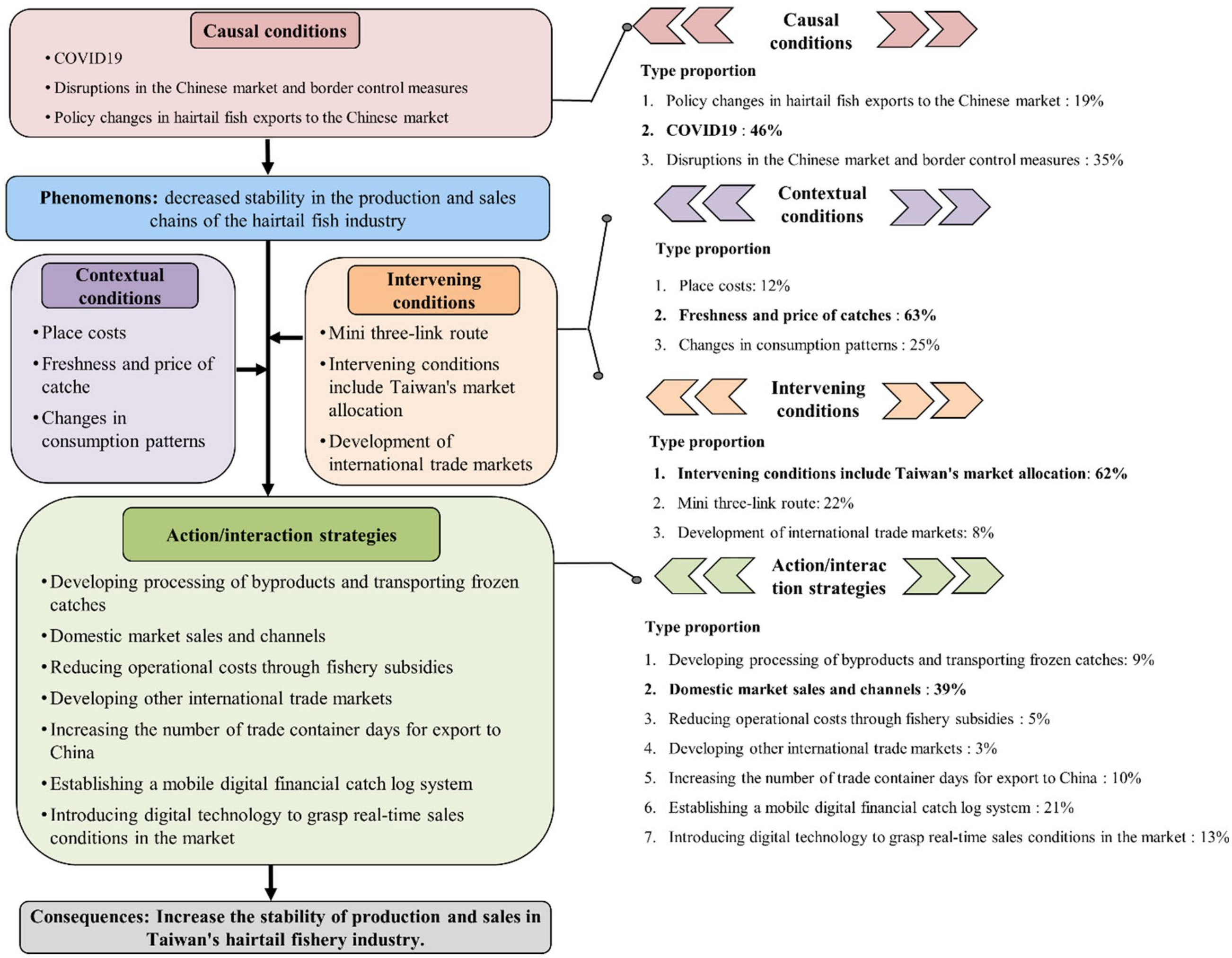

Fig. 6 The Grounded Theory analysis results on the stability of Taiwan's hairtail industry chain under COVID-19

In the case of the hairtail fishery industry in Taiwan, at the fishery production and operation levels, due to the lack of awareness of the introduction of technology by fishery producers, current management-related information is still based mainly on traditional paper records and manual operations. Therefore, since the COVID-19 pandemic began in December 2019, it has been difficult for producers to improve industry resilience due to the shortage of catch records in the hairtail fishery industry in Taiwan. Indeed, the digital capability problem that producers face is not limited to the hairtail fishery industry in Taiwan; it is a common problem faced by fisheries in many countries worldwide [10]. Although some businesses in the hairtail fishery industry have begun considering the introduction of digital technology to assist in production information records, most primary producers do not have the relevant knowledge to use the computer systems, information communication tools or digital technology to improve the efficiency of their work [18]. They also do not clearly understand how to facilitate transformation and what stages require transformation. As the investment cost of information technology is not low, the willingness of businesses to invest in the digital transformation of the industry is low.

With the increase in the importance of sustainable development in the fishing industry, food security, and environmental sustainability, advancements in digital technology could enhance the competitiveness of fishery producers. Through the collection, storage, analysis, and sharing of data, transaction costs between sellers and buyers could be effectively reduced, information transmission gaps and market barriers could be minimized, and precise and real-time policy decisions could be facilitated (World Bank Group, 2019). This is particularly significant due to the COVID-19 pandemic, which has heightened the need for contactless transactions, thereby promoting the rapid development of digital technology and market digitization in the fishing industry (Ferrer et al., 2021; Stone et al., 2020). For instance, Timor-Leste, which heavily relies on fishery products for its domestic animal protein supply, has been adopting digital transformation strategies in its fishery policies and management measures since 2017. This approach aims to increase the total catch and fish size, enhance the digital development of information related to fishing locations, and enhance local fishery producers' digitalization capabilities.

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To effectively manage the complex issues and interrelated impacts that the marine fishery industry faces, the industry may need to undergo rapid transformation, industrial innovation, and development. These changes may enhance the industry's profitability and lead to sustainable industry operations [10, 18]. In the future, the digital transformation strategy on the producer side will be biased towards the technological transformation of the operational excellence level, which will strengthen the industry's existing core capabilities and achieve the goals of making fishery production information transparent and improving operational decisions by introducing technology and enhancing digital capabilities [1, 5, 10, 18]. Additionally, future digital transformation could incorporate the concept of a mobile digital fishing log by designing a "bottom-up" data collection and visualization system for the fishing production end. Therefore, producers can quickly and conveniently log data related to fishing production, such as the catch volume, fish size, catch species, and fishing area location. Moreover, training courses on digital tools should be provided, and the importance of digital fishing logs should be promoted to fishery producers to enhance both their awareness of catch data and their acceptance of digital tools.

4 Conclusions

The hairtail fishery industry in Taiwan has experienced significant fluctuations over recent years, shaped by global demand, unforeseen challenges, and advancements in technology. The surge in production was primarily driven by increasing demand and export opportunities from China, a major consumer of seafood. China's expanding middle class and preference for high-quality fishery products spurred unprecedented growth in the hairtail market, incentivizing producers to scale up operations and optimize supply chains to meet the rising demand.

However, the COVID-19 pandemic disrupted this upward trajectory. Global lockdowns, logistical challenges, and reduced consumer purchasing power led to a sharp decline in exports and market activity. The pandemic exposed vulnerabilities in the industry, including over-reliance on single export markets and traditional distribution channels. This period of contraction highlighted the need for resilience and adaptability in the face of global crises.

Digitalization emerged as a pivotal solution during this challenging period. The integration of digital tools and platforms enabled the industry to address inefficiencies and adapt to the shifting landscape. Specifically, digital systems facilitated the even redistribution of returned products from the export market, minimizing waste and ensuring that surplus supply reached alternative markets or local consumers. By leveraging digital platforms, producers and distributors improved their inventory management, reduced losses, and maintained some level of market stability during uncertain times.

Looking ahead, continual digitalization offers the hairtail fishery industry in Taiwan a pathway to enhanced competitiveness and resilience. By adopting advanced technologies such as data analytics, blockchain for traceability, and AI-driven demand forecasting, the industry can optimize production, reduce waste, and respond swiftly to market changes. Furthermore, digital marketing and e-commerce platforms provide direct access to consumers, diversifying revenue streams and reducing dependency on traditional export markets.

In conclusion, while the hairtail fishery market faced significant challenges, digitalization has proven instrumental in fostering recovery and growth. By embracing ongoing technological advancements, the industry can not only solidify its position in global markets but also build a robust framework capable of withstanding future disruptions, ensuring sustainable and equitable growth for all stakeholders.

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Data availability The datasets generated by the analyzed during the current study are available in the FA.MOA repository (Chinese), https://www.fa.gov.tw/list.php?theme=FS_AR&subtheme.

Declarations

Ethics approval and consent to participate The study was conducted according to the guidelines of the Declaration of Helsinki and approved by the ethics committee of the Center for Taiwan Academic Research Ethics Education (No. P106050221). And, informed consent was obtained from all individual participants included in the study.

Consent to publication Not applicable.

Competing interests The authors declare no competing interests.

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